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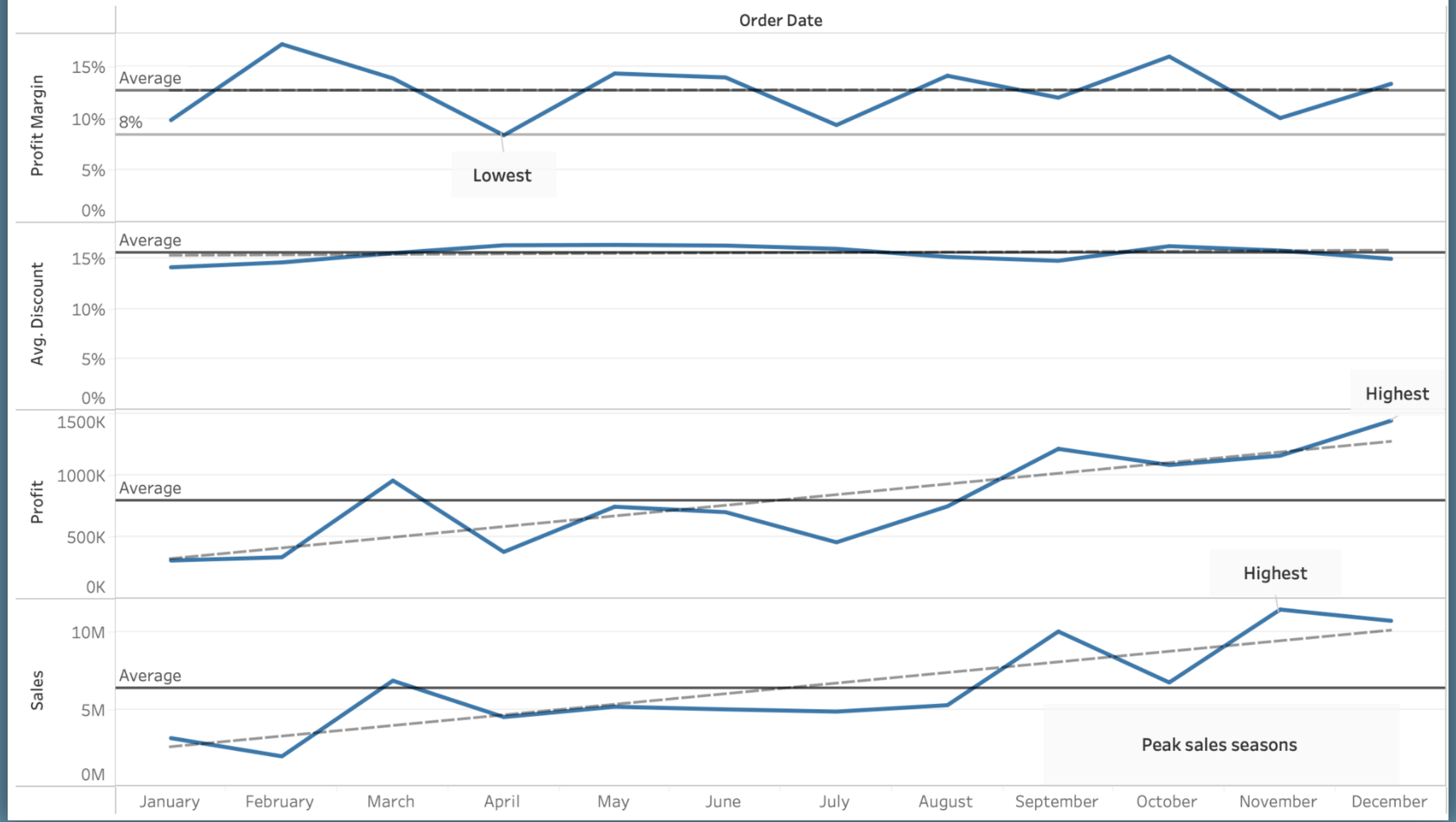
SIM-LSE DATA ANALYTICS CHALLENGE



Team Cheesy Wings

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Revenue Peaks in Q4 Without Increasing Discounts, Suggesting Seasonal Demand Drives Growth



Business Performance

- **Consistent Year-Over-Year Growth:** Both Sales and Profits exhibit a robust upward trajectory throughout the year (indicated by the rising trendlines).
- **Q4 Seasonal Surge:** Performance peaks in Q4, with sales exceeding $10M in November and profit reaching approximately $1.5M in December.
- **Organic, Stable Expansion:** The average discount rate remains stable at 15–16%, suggesting that Q4 growth is driven primarily by seasonal demand rather than more aggressive promotions.
- **Business Implication**: Despite record-breaking sales, profitability deteriorates during the company's strongest sales period, indicating that revenue growth is masking underlying structural issues.

Problem statement

THE MARGIN PARADOX:

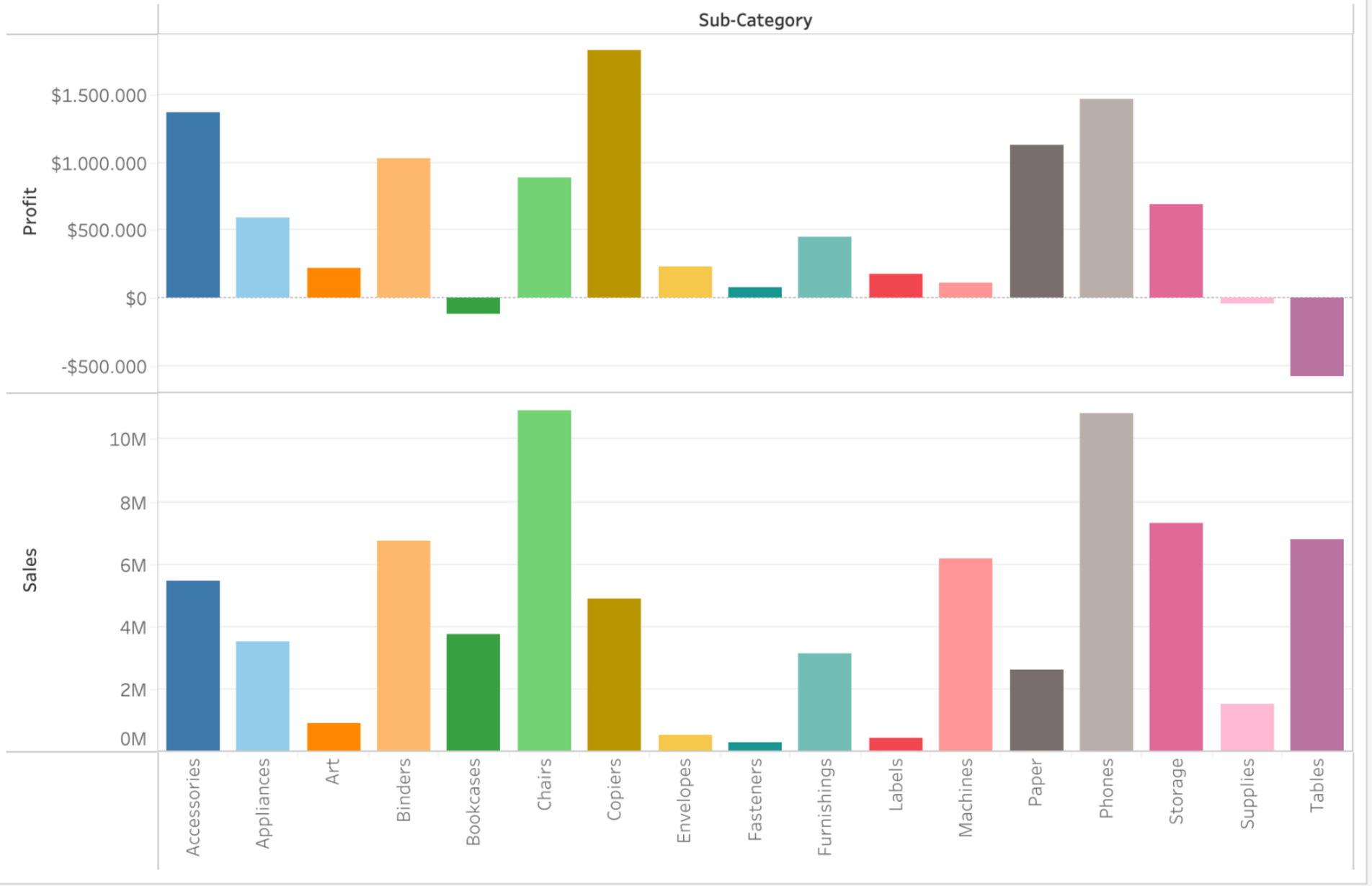
- March: March achieved the highest profit margin (~14%) despite only moderate sales.
- November: In contrast, November recorded peak sales exceeding $10 million, yet its profit margin declined sharply to around 10%.
- **Key Insight:**
- Despite a relatively stable average discount rate of 15–16%, profit margin fell from 14% in March to 10% in November, even as sales reached their annual peak.
- **This indicates that stronger sales performance did not translate into higher profitability, suggesting underlying structural inefficiencies that require further investigation.**



What is driving this margin paradox?

The Revenue-Profit Disconnect: High Volume >< High Return

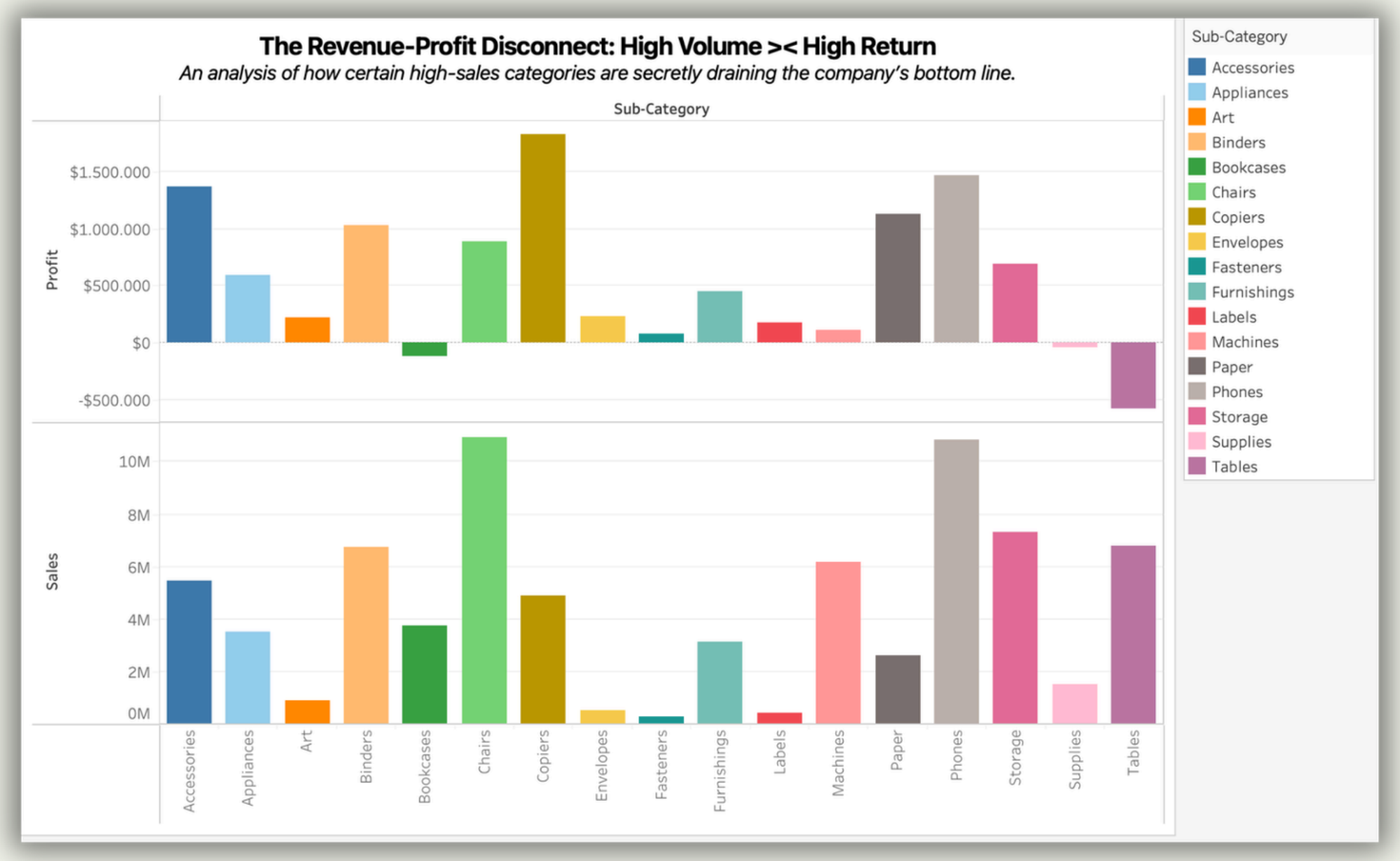
An analysis of how certain high-sales categories are secretly draining the company's bottom line.



Revenue-profit Disconnect: High volume > < High return

Notice:

- Tables though have high sales (nearly 7M),
- Tables' profits are negative and one of the lowest among other sub-categories (over -500K)
- Tables illustrate that higher sales do not mean that there would be higher profits.
- **Volume-Profit Mismatch:** Sub-categories like Binders and Chairs drive massive sales volume, yet their profit returns do not scale proportionally.
- **The Efficiency Gap:** In sharp contrast, Copiers prove to be highly efficient profit engines, generating maximum returns with much lower sales volumes.

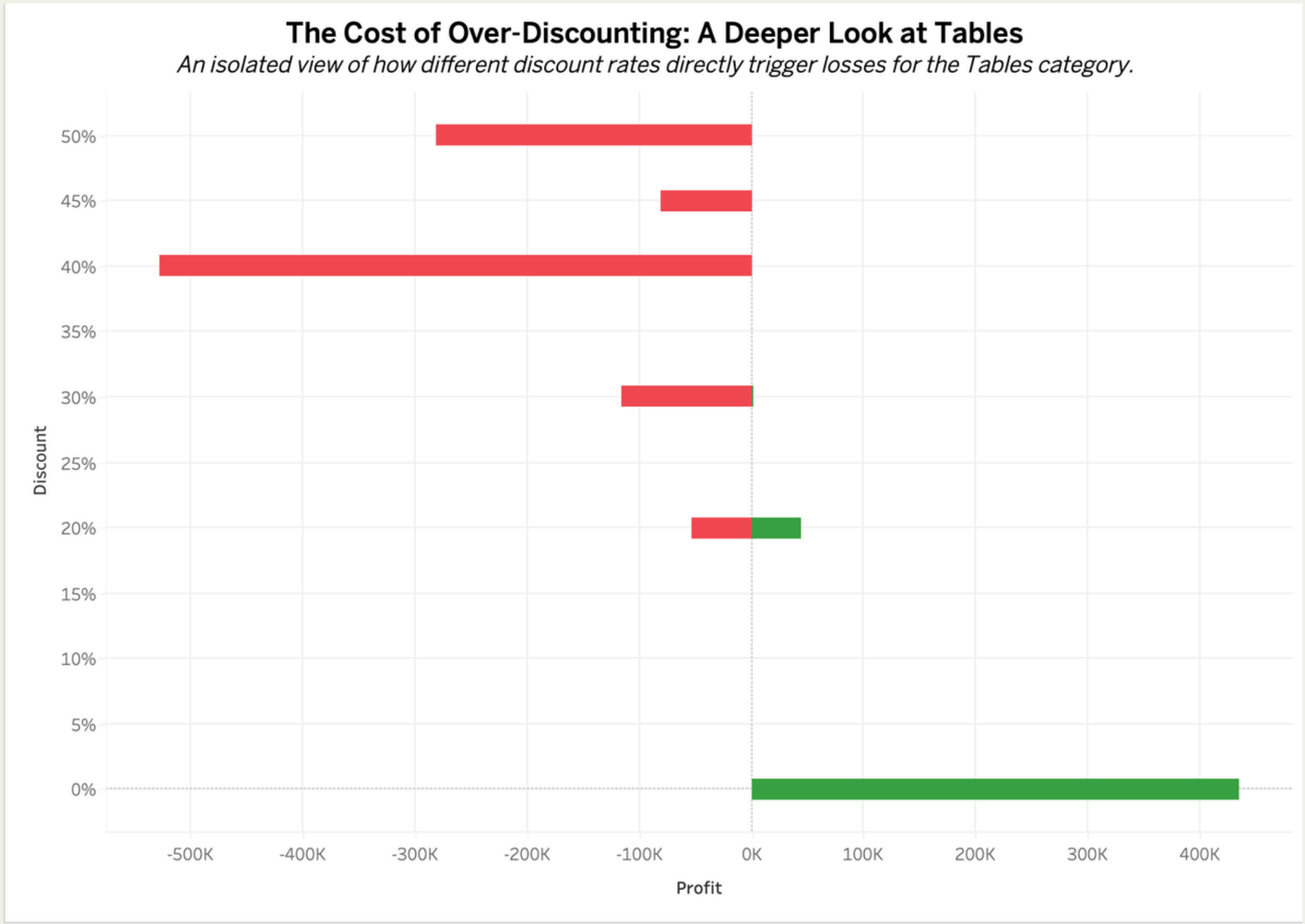


The Cost of Over-Discounting: A Deeper Look at Tables

An isolated view of how different discount rates directly trigger losses for the Tables category.



The Danger Zone: When Discounts Destroy Product Value



An isolated analysis on Tables proving the 25% threshold as a structural turning point.

Selling Tables at full price generates a solid profit of over 400K.

A 20% discount acts as the dangerous transition point, where the first wave of unprofitable transactions appears.

Any discount at or above 30% completely destroys the bottom line, driving heavy losses -500K at the 40% discount mark).

To see how these high-discount orders are distributed individually across different clients, let's look at the Scatter Plot.

The Bleeding Points: Inside the 25% Threshold Crisis

A transaction-level scatter analysis on Tables proving that excessive discounts are systematically destroying values.



Granular Analysis: Mapping the Bleeding Point of Discount Excess

The Structural Pivot (25% Threshold):

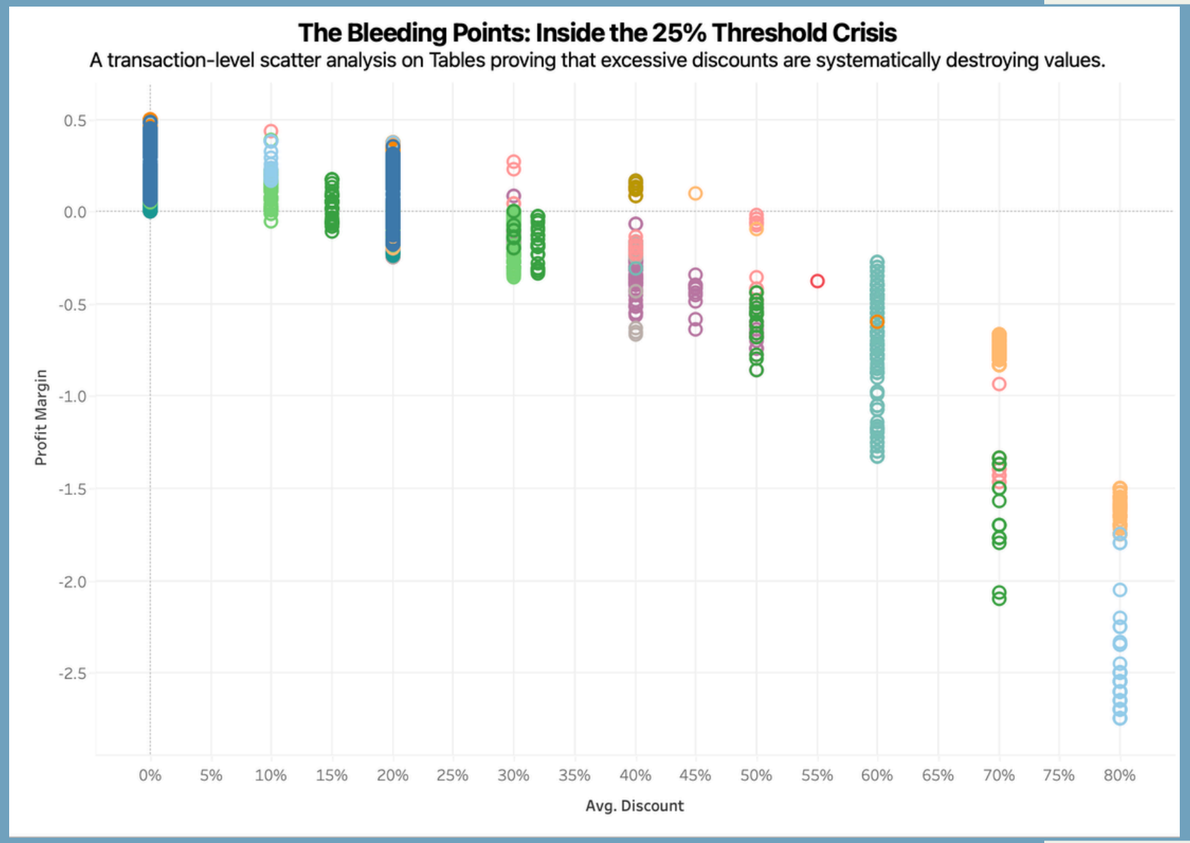
- The 25% discount mark is not just a random number; it is a structural turning point for our pricing strategy.
- Below this threshold, the vast majority of transactions stay securely in the positive territory.

The Bleeding Cluster (The Red Zone):

- Once discounts exceed 25%, we see a massive, dense cluster of orders bleeding cash, plunging heavily into negative profit margins (some reaching as low as -250%.
- This density proves that our profit loss is driven by a high volume of aggressively discounted orders, not just a few isolated accidents.

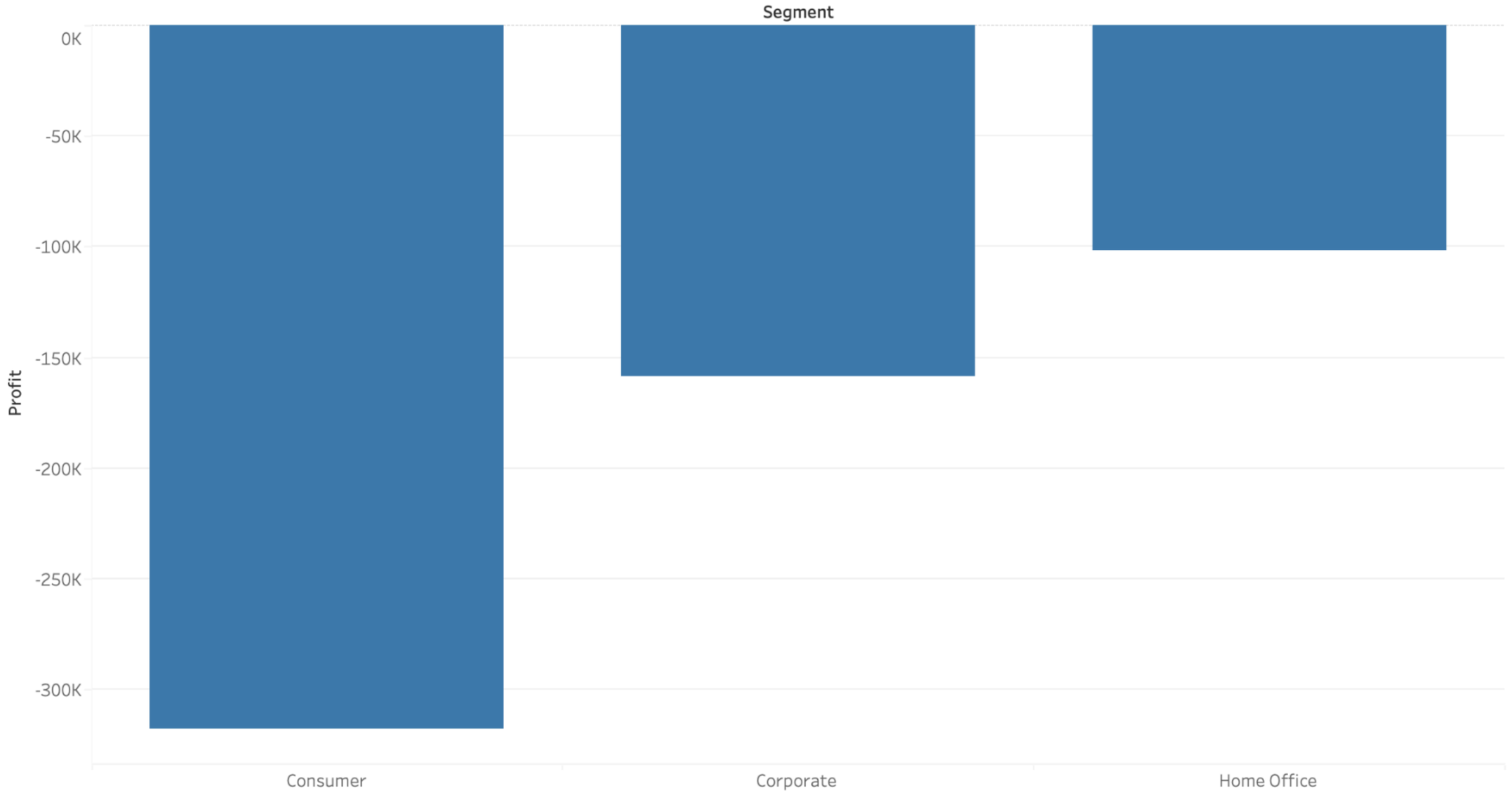
The Strategic Implication:

- To stop the bleeding, the business must implement a hard ceiling on discounts for the Tables category.
- Enforcing a strict price control policy capped at 25% would instantly eliminate these toxic, loss-making transactions without hurting our baseline sales.



Net Margin Collapse: Universal Deficit Across All Segments

A consolidated view of Tables proving that no customer segment is currently profitable, led by a massive retail sinkhole.



Universal Profit Deficit Across All Customer Segments

1 Universal Profit Deficit

Not a single customer segment turns a profit for the Tables category. The financial bleeding is a structural, system-wide issue.

2 The Consumer Retail Sinkhole

The Consumer segment is the absolute heaviest profit drainer, pulling in a staggering net loss of over -$300K.

3 Corporate Negotiation Failure

Even the bulk-buying business channel (Corporate) failed to break even, caving into a massive deficit of nearly -$160K.

4 Home Office Cap Collapse

Small/Home Office accounts also plunged deep into the negative territory at around -$100K.

5 Aggressive Discount Destructive Nature

The uniform collapse confirms that our historic sales volume did not fail due to customer preferences, but because aggressive discounting structurally ruined product value across all consumer types.

Overall Recommendations

Enforce a Strict 25% Discount Ceiling: Lock system-level discounts for Tables at a maximum of 25% to instantly eliminate toxic, loss-making transactions.

De-risk the Consumer Channel: Shift the Consumer retail channel from flat cash discounts to non-monetary value additions like free shipping or product bundles.

Mandate Corporate Floor Prices: Introduce strict minimum floor margins for bulk Corporate contracts to stop unmonitored profit leakage during sales negotiations.

Capitalize on Q4 Seasonal Demand: Leverage organic Q4 peak sales seasons to maximize full-price volume instead of relying on aggressive promotion campaigns.

Optimize Product Portfolio Mix: Reallocate marketing budgets away from structural loss-making items (Tables) toward high-performing, profitable product sub-categories.